Wasserstein–Fisher–Rao Gaussian Gradient Flow: Phase Separation Between Transport and Natural-Gradient Descent

Abstract

We study the Wasserstein–Fisher–Rao (WFR) gradient flow on the Gaussian variational manifold, a hybrid of the Bures–Wasserstein transport flow and the Fisher–Rao (natural-gradient) flow with a tunable transport strength λ_t. We discretize the flow by forward–backward operator splitting – one Wasserstein half-step followed by one Fisher–Rao half-step per iteration – and compare it against the two pure flows ($\lambda \equiv 0$, Fisher–Rao only; and Wasserstein only) on two ill-conditioned two-dimensional targets: an exact anisotropic Gaussian posterior and a smooth nonseparable strongly log-concave posterior. From a far, underdispersed initialization the experiments expose a clean *phase separation*: the Wasserstein transport drives fast early descent (warmup) but stalls in the tail, while the Fisher–Rao step is slow during warmup but converges rapidly once the covariance is locally calibrated to the curvature. The WFR splitting inherits both phases, and a curvature-adaptive transport schedule $h_n = h_{\max}/(1 + (s_n/s_0)^2)$ – large transport when the state is underdispersed, decaying transport as the covariance calibrates – realizes the hand-off automatically.

Cost matters, and the honest accounting is nuanced. Per **iteration**, the WFR splitting reaches every tolerance in roughly half the iterations of pure Fisher–Rao. But the splitting evaluates the posterior expectations *twice* per iteration (once at each half-step), so on the fair **expectation-batch** axis the per-iteration advantage is largely cancelled: to tight tolerances pure Fisher–Rao is the most batch-efficient method, the WFR splitting is competitive (and overtakes it on the loosest warmup threshold in the hardest, most underdispersed regimes), and Wasserstein-only is decisively worst – it plateaus and never reaches 10^{-6} at $\Lambda = 1000$. The adaptive schedule dominates the fixed and theorem-bound schedules among WFR variants; the theorem bound $h = \mu_{\min}$ is far too conservative once the state is underdispersed.

Contents

1 Setup

Gaussian variational family and energy. We approximate a target posterior $\rho_{\text{post}} \propto e^{-V}$ on $\mathbb{R}^d$ ($d = 2$ throughout) by a Gaussian $\mathcal{N}(m, C)$, $C \in \text{SPD}(d)$, parameterized by the state $a = (m, C)$. The variational objective is the (reverse) KL up to an a-independent constant,

$$\mathcal{E}(a) \;=\; \mathbb{E}_{\theta\sim\mathcal{N}(m,C)}[V(\theta)] \;-\; \tfrac{1}{2}\log\det C, \qquad \mathcal{E}(a) - \mathcal{E}(a_\star) \;=\; \mathrm{KL}\big(\mathcal{N}(m, C) \,\big\|\, \rho_{\text{post}}\big), \tag{1}$$

minimized at $a_\star = (m_\star, C_\star)$. We write the posterior log-gradient and log-Hessian expectations

$$g \;=\; \mathbb{E}_{\mathcal{N}(m,C)}[\nabla \log \rho_{\text{post}}] \;=\; -\,\mathbb{E}[\nabla V], \qquad H \;=\; \mathbb{E}_{\mathcal{N}(m,C)}[\nabla^2 \log \rho_{\text{post}}] \;=\; -\,\mathbb{E}[\nabla^2 V], \tag{2}$$

so that $-H = \mathbb{E}[\nabla^2 V] \succeq 0$ for a log-concave target. We assume global strong log-concavity and smoothness, $\alpha I \preceq \nabla^2 V \preceq \beta I$.

The WFR Gaussian flow. The Bures–Wasserstein gradient flow of $\mathcal{E}$ moves mass by transport; the Fisher–Rao flow rescales mass by the natural gradient. The WFR flow combines them with transport strength $\lambda_t \geq 0$:

$$\dot{m} \;=\; (C + \lambda_t I)\, g, \qquad \dot{C} \;=\; \underbrace{C + CHC}_{\text{Fisher–Rao}} + \lambda_t \underbrace{\big(2I + CH + HC\big)}_{\text{Wasserstein}}. \tag{3}$$

Setting $\lambda_t \equiv 0$ recovers the Fisher–Rao (natural-gradient) flow; dropping the Fisher–Rao terms recovers the Bures–Wasserstein flow. The additive $2\lambda_t I$ term is the source of the warmup advantage analyzed below: it inflates an underdispersed covariance at a rate independent of the current scale.

Targets. The anisotropy parameter is Λ (the symbol λ is reserved for the transport strength λ_t). We use two targets.

- **Gaussian posterior:** $\rho_{\text{post}} = \mathcal{N}(0, \Gamma)$ with $\Gamma = \text{diag}(1, \Lambda)$ the *covariance*. Then $V(\theta) = \frac{1}{2}\theta^\top \Gamma^{-1}\theta$, so $g = -\Gamma^{-1} m$ and $H = -\Gamma^{-1}$ are exact and state-independent, $\alpha = 1/\Lambda$, $\beta = 1$, and the optimum is closed form $m_\star = 0$, $C_\star = \Gamma$ with an analytic energy gap (1).
- **Smooth log-cosh posterior:** the nonseparable strongly log-concave

$$V(x, y) \;=\; \tfrac{1}{20}\big(\sqrt{\Lambda}\, x - y\big)^2 \;+\; \tfrac{\delta}{2}\,(x^2 + y^2) \;+\; \gamma\, \log\cosh(y), \quad \delta = 0.05,\ \gamma = 1, \tag{4}$$

  with $\alpha = \delta$ and the eigenvalue smoothness constant $\beta = \lambda_{\max}\big(\begin{smallmatrix} \delta+\Lambda/10 & -\sqrt{\Lambda}/10 \\ -\sqrt{\Lambda}/10 & \delta+1/10+\gamma \end{smallmatrix}\big)$. Expectations of tanh, sech^2 and $\log\cosh$ under the marginal $Y \sim \mathcal{N}(m_y, C_{22})$ are evaluated by deterministic Gauss–Hermite quadrature; the optimum is found numerically.

The target metadata, including β/α (the condition number that governs the tail rate), is reported in Table 1.

Initialization. Every run starts far and underdispersed: $C_0 = \varepsilon I$ and the mean placed a distance $r = 8$ along the flat-valley direction $(1, \sqrt{\Lambda})/\sqrt{1 + \Lambda}$. This is the regime that separates the two flows – the covariance must inflate by orders of magnitude and the mean must traverse the valley.

2 Forward–backward discretization and the five methods

Operator splitting. We discretize (3) by forward–backward operator splitting with Fisher–Rao step Δt and Wasserstein step $h_n = \lambda_n \Delta t$. Each iteration applies one Wasserstein half-step then one Fisher–Rao half-step. With g_n, H_n from (2) evaluated at $a_n = (m_n, C_n)$, the *Wasserstein half-step* is

$$\begin{aligned} &m_{n+1/2} = m_n + h_n\, g_n, \qquad\qquad M_n = I + h_n H_n, \qquad \widetilde{C} = M_n C_n M_n, \\ &C_{n+1/2} = \tfrac{1}{2}\Big(\widetilde{C} + 2h_n I + \big[\widetilde{C}(\widetilde{C} + 4h_n I)\big]^{1/2}\Big), \end{aligned} \tag{5}$$

and, with $g_{n+1/2}, H_{n+1/2}$ re-evaluated at $a_{n+1/2}$, the *Fisher–Rao half-step* is the KL/natural-gradient update

$$m_{n+1} = m_{n+1/2} + \Delta t\, C_{n+1/2}\, g_{n+1/2}, \qquad C_{n+1} = (1 + \Delta t)\big(C_{n+1/2}^{-1} - \Delta t\, H_{n+1/2}\big)^{-1}. \tag{6}$$

The matrix square root in (5) is computed spectrally: $\widetilde{C}$ and $\widetilde{C} + 4h_n I$ commute, so with the symmetric eigendecomposition $\widetilde{C} = V \operatorname{diag}(w)\, V^\top$,

$$C_{n+1/2} = V \operatorname{diag}\Big(\tfrac{1}{2}\big(w + 2h_n + \sqrt{w(w + 4h_n)}\big)\Big)\, V^\top, \tag{7}$$

which is exact, symmetric, and SPD whenever $\widetilde{C} \succeq 0$ and $h_n \geq 0$. The Fisher–Rao covariance update (6) is unconditionally SPD for log-concave targets ($-H \succeq 0$). At $h_n = 0$ the Wasserstein half-step is the identity and the iteration reduces exactly to the pure Fisher–Rao step.

Cost: expectation batches. The fair cost axis is the number of *expectation batches* (evaluations of (g, H)): the single-step methods spend one per iteration, the full WFR splitting spends two (one at a_n, one at $a_{n+1/2}$). We report convergence against expectation batches throughout, so the WFR splitting is charged for its second evaluation.

The five methods.

- **FR-only** ($\lambda \equiv 0$): pure Fisher–Rao, $h_n = 0$.
- **W-only**: the Wasserstein half-step (5) alone with the fixed $h_n = c/\beta$.
- **WFR-fixed**: the full splitting with fixed $h_n = c/\beta$.
- **WFR-theory**: the full splitting with the constant $h_n = \mu_{\min} = \min(\lambda_{\min}(C_0), 1/\beta)$, the maximizer of the proven discrete transport contribution $f(h) = \alpha h/(1 + h/\mu_{\min})^2$.
- **WFR-adaptive**: the full splitting with the curvature-adaptive schedule

$$h_n = \frac{h_{\max}}{1 + (s_n/s_0)^2}, \quad h_{\max} = \frac{0.9}{\beta}, \quad s_0 = 0.5, \quad s_n = \lambda_{\min}\big(C_n^{1/2}(-H_n)\, C_n^{1/2}\big). \tag{8}$$

The calibration ratio s_n is the smallest eigenvalue of the dimensionless curvature-in-covariance-metric operator: $s_n \ll 1$ signals an underdispersed state (the Fisher–Rao step is throttled, large transport helps warmup), while $s_n \approx 1$ signals a locally calibrated covariance (the Fisher–Rao step is reliable, transport should decay). For the Gaussian target at $C = \Gamma$, $-H = \Gamma^{-1}$ gives $C^{1/2}(-H)C^{1/2} = I$ and $s_n = 1$ exactly.

Table 1: Target metadata: strong-log-concavity α, smoothness β, condition number β/α, initial covariance scale ε, and the initial energy gap $\text{gap}_0 = \mathcal{E}(a_0) - \mathcal{E}(a_\star)$.

target	Λ	ε	α	β	β/α	gap_0
Gaussian	100	0.001	0.010	1.000	100.0	8.8
Gaussian	100	0.01	0.010	1.000	100.0	6.5
Gaussian	1000	0.001	0.001	1.000	1000.0	9.4
Gaussian	1000	0.01	0.001	1.000	1000.0	7.1
smooth log-cosh	100	0.001	0.050	10.161	203.2	14
smooth log-cosh	100	0.01	0.050	10.161	203.2	12
smooth log-cosh	1000	0.001	0.050	100.151	2003.0	13
smooth log-cosh	1000	0.01	0.050	100.151	2003.0	11

3 Results

Unless stated otherwise the head-to-head uses a shared Fisher–Rao step $\Delta t = 0.5$ and the fixed fraction $c = 0.5$ for W-only and WFR-fixed, on the production grid $\Lambda \in \{100, 1000\}$, $\varepsilon \in \{10^{-2}, 10^{-3}\}$. Target constants are in Table 1.

3.1 Phase separation: transport wins warmup, Fisher–Rao wins the tail

Figures 1–2 show the energy gap against expectation batches, with a warmup zoom (left) and the full trajectory (right). Three facts are consistent across every (Λ, ε):

1. **W-only descends fastest in the first few batches** but *plateaus*: from $C_0 = \varepsilon I$ the additive $2h_n I$ term in (5) inflates the covariance at a scale-independent rate, collapsing the bulk of the initial gap quickly, after which the lack of the natural-gradient rescaling leaves it stalled far from $a_\star$.

2. **FR-only is slow during warmup** – the natural-gradient covariance inflates only geometrically and the mean update $\Delta t\, Cg$ is throttled by the tiny C – **but converges fastest in the tail** once the covariance is calibrated.

3. **WFR (fixed and adaptive) inherit both phases**: they track W-only's fast warmup and then FR-only's fast tail. *Per iteration* this makes them the fastest methods – the adaptive schedule reaches gap $< 10^{-3}$ in about half the iterations of FR-only (Table 2). The expectation-batch cost of that per-iteration advantage is the subject of Section 3.4.

3.2 The mechanism: the covariance, not the mean, is the bottleneck

The phase separation is not just a property of the scalar energy gap – it has a concrete mechanism, visible once the gap is decomposed into its mean and covariance parts. Figure 3 plots $\|m_n - m_\star\|$ and $\|C_n - C_\star\|_F$ separately on the hardest start ($\Lambda = 1000$, $\varepsilon = 10^{-3}$). Three observations pin down why the flows behave as they do:

1. **The covariance dominates the error budget.** At the underdispersed start the covariance error is $\approx 10^3$ (it must inflate from εI to a top eigenvalue $\Lambda = 10^3$) while the mean error is only ≈ 8. The optimization is, to leading order, a covariance-inflation problem; the mean is a sideshow.

2. **The mean always calibrates before the covariance.** For every convergent method the mean error crosses 10^{-3} tens of batches before the covariance error does (e.g. on the smooth target, FR-only at 39 vs 52 batches; WFR-adaptive at 48 vs 84). The tail rate is therefore set entirely by how fast the covariance is calibrated – which is exactly the quantity the Fisher–Rao step accelerates and the adaptive ratio s_n measures.

3. **W-only inflates volume but cannot calibrate shape.** Its covariance error barely moves (it drifts from $\approx 10^3$ down to $\approx 8.8 \times 10^2$ over the first 128 batches) because the additive $2h_n I$ term grows the volume isotropically without ever matching the target's anisotropy. This is the mechanism behind its plateau: transport produces a big but mis-shaped covariance, and only the natural-gradient rescaling can fix the shape.

3.3 The adaptive schedule tracks curvature calibration

Figure 4 shows the normalized transport $h_n\beta$ and the calibration ratio s_n along the WFR-adaptive runs. The schedule holds transport near the ceiling $h_{\max}\beta = 0.9$ while $s_n \ll s_0$ (the underdispersed warmup) and decays it as s_n approaches 1 (the covariance becomes locally calibrated and the Fisher–Rao step takes over), exactly the hand-off implied by the phase separation.

3.4 Cost accounting: iterations versus expectation batches

The phase figures already plot the fair cost axis. We now make the trade-off explicit. Table 2 reports, for the harder underdispersed start $\varepsilon = 10^{-3}$, the iterations and the expectation batches each method needs to reach gap $< 10^{-3}$.

- **Per iteration the WFR splitting wins.** WFR-adaptive reaches the tolerance in 15–22 iterations against FR-only's 30–41 – roughly half, because each iteration blends a transport move and a natural-gradient move.

- **Per expectation batch the picture reverses.** The full splitting evaluates (g, H) twice per iteration (once at a_n, once at $a_{n+1/2}$), so its batch cost is about double its iteration count, whereas FR-only spends one batch per iteration. Once charged this way, *FR-only is the most batch-efficient method to the tight tolerances* (30–41 batches against WFR-adaptive's 42–58). WFR-adaptive remains competitive and is never far behind; it *overtakes* FR-only only on the loosest, warmup-dominated threshold 10^{-1} in the hardest regimes (Table 3), where its faster warmup outruns the doubled per-iteration cost before the Fisher–Rao tail takes over.

- **W-only is decisively worst.** It plateaus far from the optimum and never reaches 10^{-6} at $\Lambda = 1000$ within the batch budget (Table 3); transport alone cannot calibrate the covariance.

All twenty production runs per method are SPD-feasible and monotone throughout.

3.5 How much transport? The schedule sweep

Figure 5 sweeps the fixed fraction c (so $h = c/\beta$, with $c = 0$ recovering FR-only) on both cost axes; Table 4 gives the batch numbers. The two rows of the figure tell the trade-off cleanly: increasing c *monotonically reduces the iteration count* (top), but the expectation-batch cost (bottom) sits *above* the $c = 0$ (FR-only) value for every positive c, because each WFR iteration costs two batches. The theorem-bound schedule $h = \mu_{\min}$ (star markers) lands in the interior of the fixed-c curve rather than at an optimum: with the tiny $C_0 = \varepsilon I$ the bound gives $\mu_{\min} = \varepsilon$ on the Gaussian target (since

Table 2: Iterations versus expectation batches to reach gap $< 10^{-3}$ at the harder start $\varepsilon = 10^{-3}$. WFR-adaptive needs the fewest iterations (about half of FR-only), but at two batches per iteration FR-only is the most batch-efficient; W-only is far behind on both axes.

		iterations		exp. batches	
target	method	Λ=100	Λ=1000	Λ=100	Λ=1000
Gaussian	W-only	575	3653	575	3653
	FR-only	36	41	36	41
	WFR-fixed	16	22	32	44
	WFR-theory	31	37	62	74
	WFR-adaptive	15	21	30	42
smooth log-cosh	W-only	260	2560	260	2560
	FR-only	31	31	31	31
	WFR-fixed	17	23	34	46
	WFR-theory	26	26	52	52
	WFR-adaptive	16	22	32	44

$\beta = 1$), an extremely small transport that throttles the warmup the bound was meant to accelerate – direct evidence that the conservative proof-bound schedule underuses transport precisely in the underdispersed regime it targets. The full five-method comparison is in Table 3.

3.6 Stability across the discretization step

The appendix heatmap (Figure 6) sweeps the Fisher–Rao step Δt against the transport fraction c for WFR-fixed, confirming that the splitting remains SPD-feasible and convergent across a broad joint range rather than at a single tuned Δt.

4 Findings and implications

1. **The two flows phase-separate.** Wasserstein transport, through its scale-independent $2h_nI$ covariance inflation, dominates the warmup; Fisher–Rao natural-gradient descent dominates the tail once the covariance is calibrated to the curvature. The separation is clean and holds across both targets and all (Λ, ε).

2. **The WFR splitting combines the phases, halving iterations.** Per iteration it reaches every tolerance in roughly half the iterations of pure Fisher–Rao (Table 2).

3. **On the fair cost axis the verdict is nuanced, not triumphant.** Because the splitting evaluates the posterior expectations twice per iteration, pure Fisher–Rao is the most expectation-batch-efficient method to tight tolerances. The WFR splitting is competitive and overtakes Fisher–Rao only on the loosest, warmup-dominated threshold in the most underdispersed regimes. The practical implication: *transport earns its second evaluation only when the warmup phase dominates the budget* – a far start, a very tight transport-free warmup bottleneck, or a loose target tolerance. When iterations rather than expectation evaluations are the scarce resource (e.g. a cheap closed-form (g, H)), the splitting is preferable outright.

4. **Adaptivity beats both fixed schedules, and the proof bound is too conservative.** Among WFR variants the curvature-adaptive schedule dominates fixed c and the theorem

bound $h = \mu_{\min}$; the latter collapses to a negligible transport in exactly the underdispersed regime it was meant to accelerate.

5. **Transport alone is not enough.** W-only plateaus and never reaches 10^{-6} at $\Lambda = 1000$: without the natural-gradient rescaling it cannot calibrate the covariance.

5 Limitations

The study is deliberately controlled and its scope bounds the claims. *(i) Dimension and targets:* $d = 2$, two globally strongly-log-concave targets; the batch/iteration trade-off constants will differ in higher dimension and the phase separation is not established for multimodal or heavy-tailed targets. *(ii) Exact expectations:* (g, H) are exact (closed form or Gauss–Hermite), so the comparison isolates the discretization, not Monte-Carlo gradient noise; under sampling noise the relative ranking on the batch axis could shift (transport's warmup robustness may matter more, or the second evaluation's variance may matter more). *(iii) First-order splitting:* we use one Wasserstein then one Fisher–Rao half-step (Lie–Trotter); a Strang splitting or a single fused step could change the per-iteration batch count and is untested. *(iv) Cost model:* "expectation batches" charges (g, H) evaluations equally and ignores the per-step linear algebra (eigendecomposition, matrix inverse), which is negligible at $d = 2$ but not in general. *(v) The adaptive schedule* is one heuristic with hand-set $h_{\max} = 0.9/\beta$, $s_0 = 0.5$; we do not claim it is optimal, only that it dominates the fixed and proof-bound schedules tested.

A Discretization-step robustness

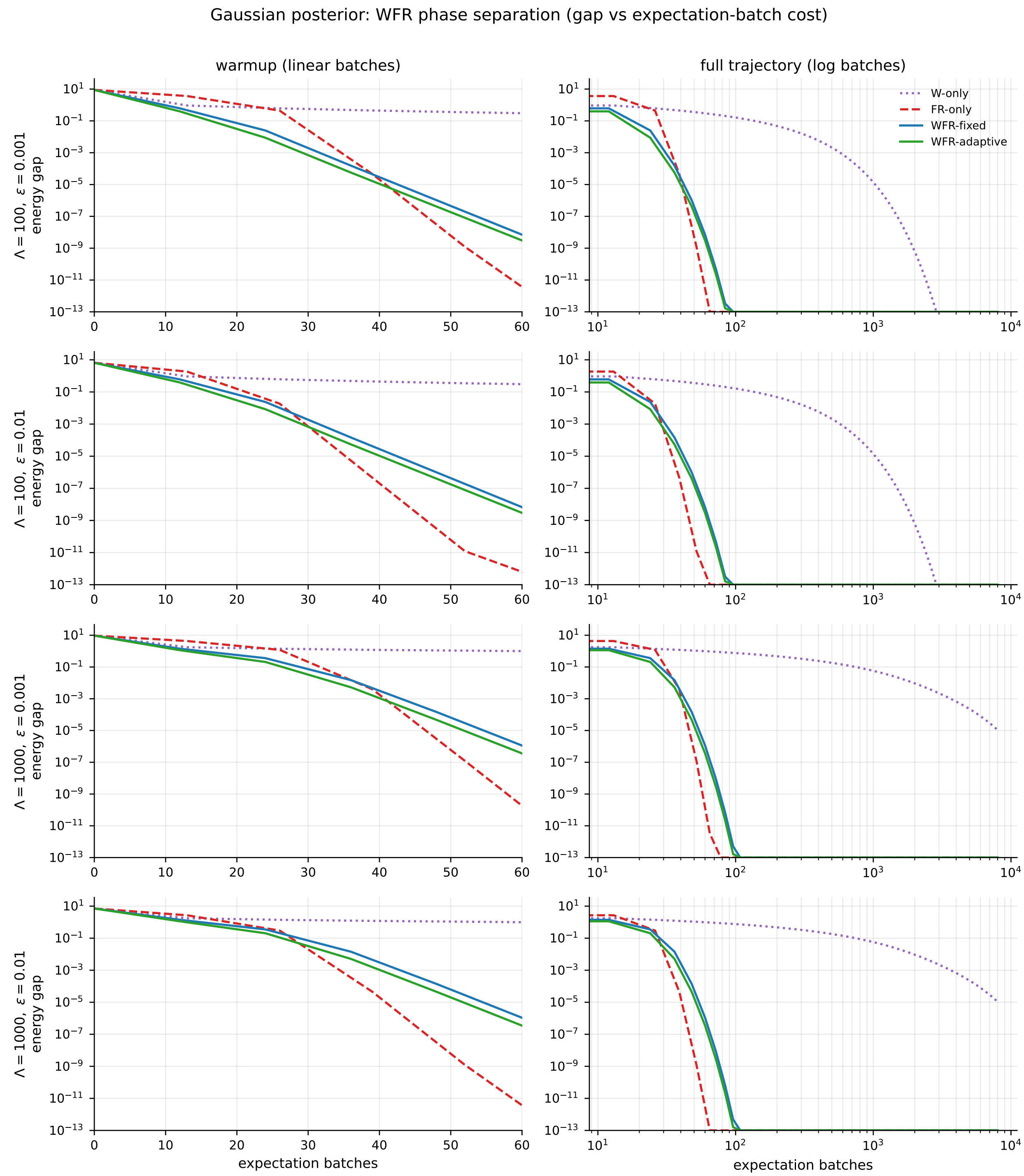

Figure 1: Gaussian posterior: energy gap vs expectation batches, warmup zoom (left) and full trajectory (right), one row per (Λ, ε). W-only wins the first batches then plateaus; FR-only lags then overtakes; WFR-adaptive captures both phases.

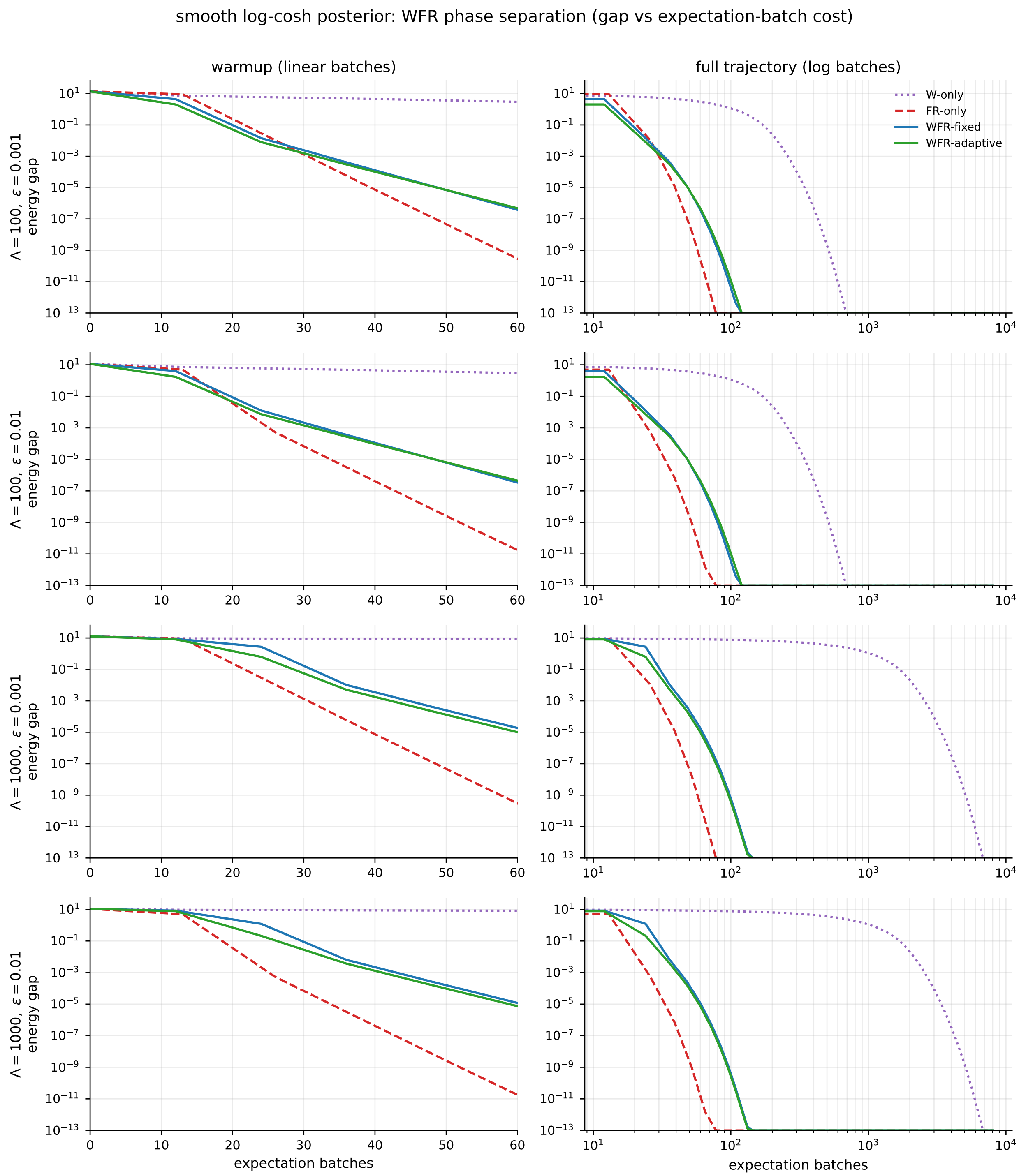

Figure 2: Smooth log-cosh posterior: same layout as Figure 1. The phase separation persists on the nonseparable, numerically-optimized target.

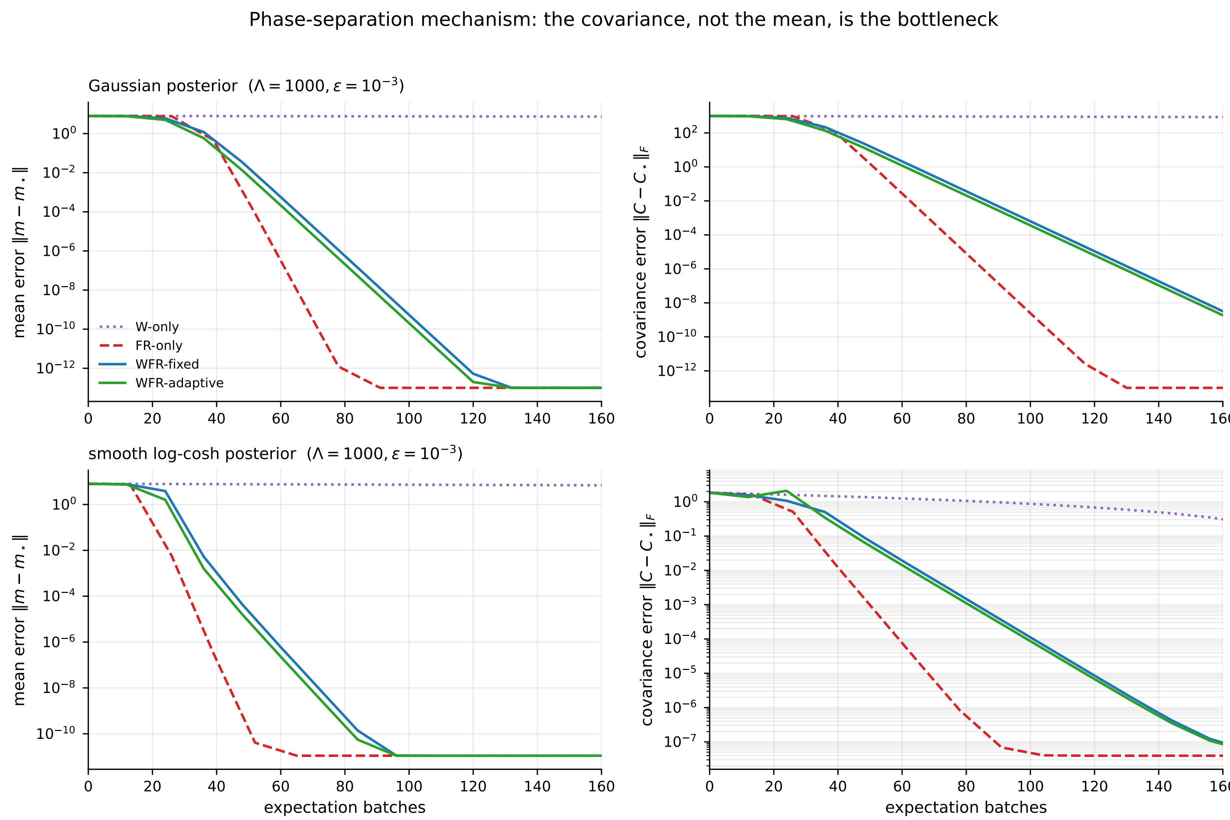

Figure 3: Phase-separation mechanism on the hardest start ($\Lambda = 1000$, $\varepsilon = 10^{-3}$): mean error $\|m - m_\star\|$ (left) and covariance error $\|C - C_\star\|_F$ (right), Gaussian (top) and smooth log-cosh (bottom), vs expectation batches. The covariance error starts two orders of magnitude above the mean error; W-only never collapses it, while FR-only and WFR do once the mean has arrived.

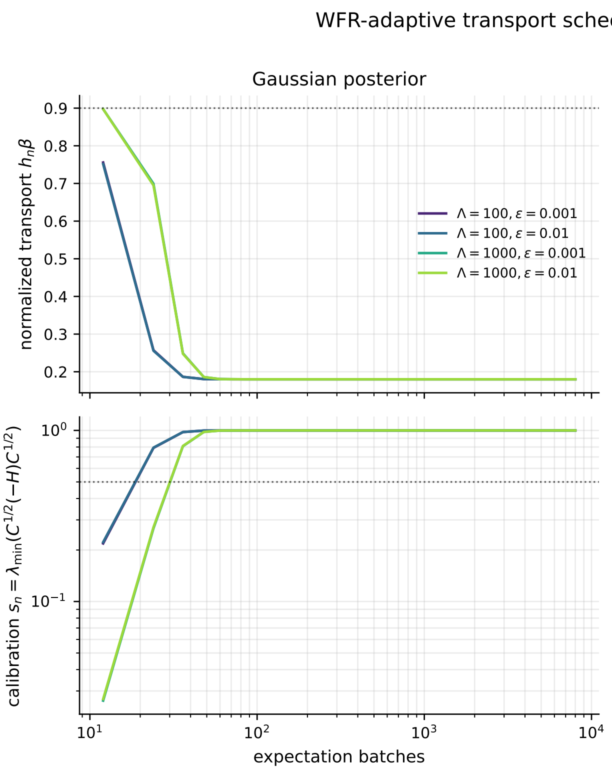

Figure 4: WFR-adaptive transport schedule. Top: normalized transport $h_n\beta$ (ceiling 0.9 dotted). Bottom: calibration ratio $s_n = \lambda_{\min}(C_n^{1/2}(-H_n)C_n^{1/2})$ (log scale, $s_0 = 0.5$ dotted). Transport stays high through warmup and decays as the state calibrates.

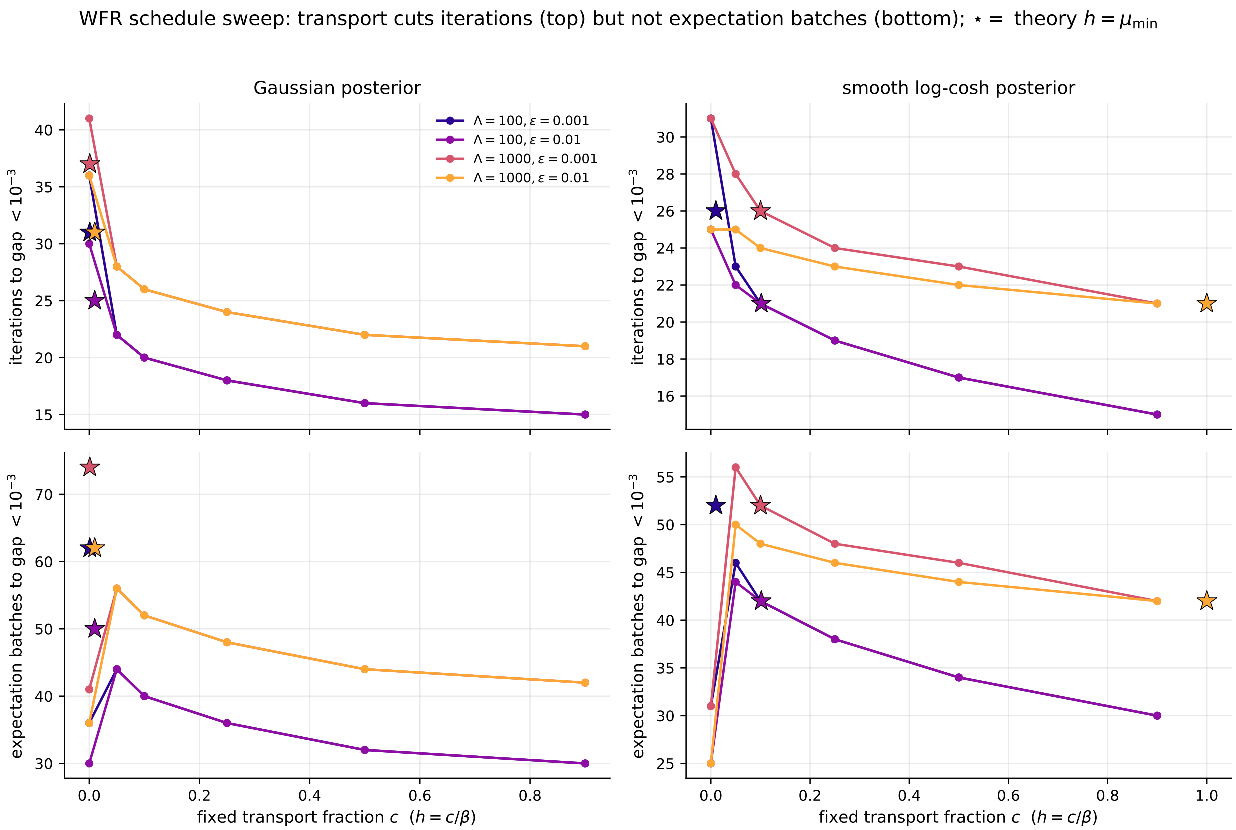

Figure 5: Schedule sweep: expectation batches to gap $< 10^{-3}$ vs the fixed transport fraction c $(h = c/\beta)$, one curve per (Λ, ε). $c = 0$ is FR-only; $\star$ marks the theory schedule $h = \mu_{\min}$.

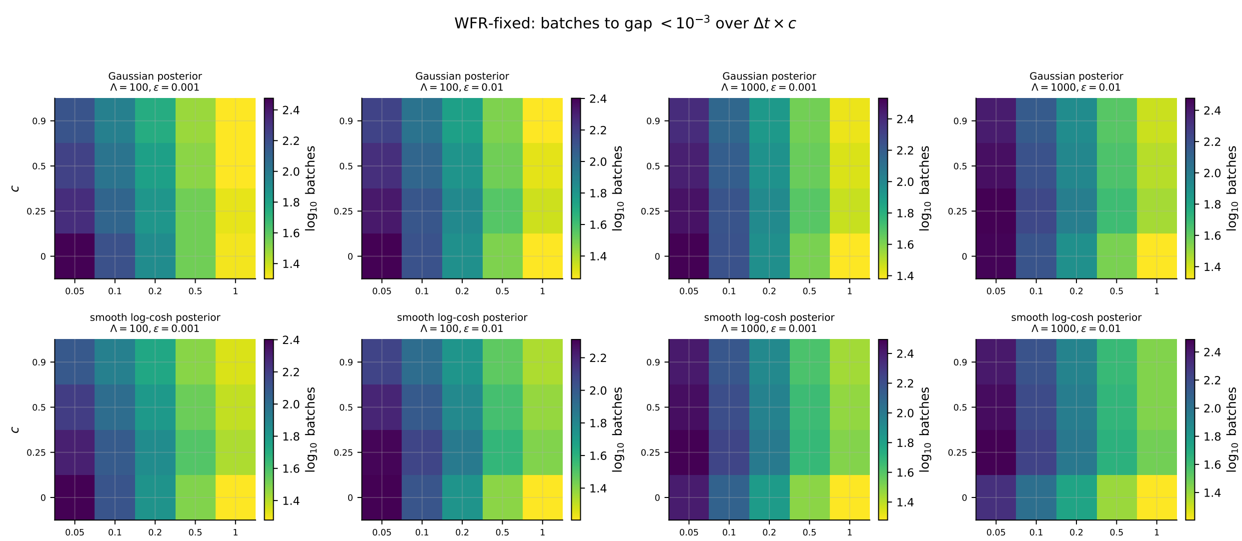

Figure 6: WFR-fixed: $\log_{10}$ expectation batches to gap $< 10^{-3}$ over the $\Delta t \times c$ grid, per target and (Λ, ε). White cells were not SPD-feasible or did not reach the threshold within budget.

Table 3: Main hitting-batch comparison: expectation batches to reach energy gap below 10^{-1}, 10^{-3}, 10^{-6}, and the final gap, per target/Λ/ε/method. ∞ denotes the threshold was not reached within the batch budget.

target	method	Λ	ε	$b_{10^{-1}}$	$b_{10^{-3}}$	$b_{10^{-6}}$	$\text{gap}_{\text{final}}$
Gaussian	FR-only	100	0.001	30	36	44	0
	W-only	100	0.001	138	575	1264	0
	WFR-adaptive	100	0.001	18	30	46	0
	WFR-fixed	100	0.001	22	32	50	0
	WFR-theory	100	0.001	50	62	80	0
Gaussian	FR-only	100	0.01	24	30	39	0
	W-only	100	0.01	138	575	1264	0
	WFR-adaptive	100	0.01	18	30	46	0
	WFR-fixed	100	0.01	20	32	48	0
	WFR-theory	100	0.01	38	50	68	0
Gaussian	FR-only	1000	0.001	35	41	50	0
	W-only	1000	0.001	746	3653	∞	1.1×10^{-5}
	WFR-adaptive	1000	0.001	28	42	58	$< 10^{-12}$
	WFR-fixed	1000	0.001	30	44	62	$< 10^{-12}$
	WFR-theory	1000	0.001	60	74	90	0
Gaussian	FR-only	1000	0.01	29	36	44	0
	W-only	1000	0.01	746	3653	∞	1.1×10^{-5}
	WFR-adaptive	1000	0.01	28	42	58	$< 10^{-12}$
	WFR-fixed	1000	0.01	30	44	62	$< 10^{-12}$
	WFR-theory	1000	0.01	48	62	80	0
smooth log-cosh	FR-only	100	0.001	23	31	44	$< 10^{-12}$
	W-only	100	0.001	171	260	387	$< 10^{-12}$
	WFR-adaptive	100	0.001	18	32	58	$< 10^{-12}$
	WFR-fixed	100	0.001	20	34	58	$< 10^{-12}$
	WFR-theory	100	0.001	36	52	80	$< 10^{-12}$
smooth log-cosh	FR-only	100	0.01	17	25	39	$< 10^{-12}$
	W-only	100	0.01	171	260	387	$< 10^{-12}$
	WFR-adaptive	100	0.01	16	32	58	$< 10^{-12}$
	WFR-fixed	100	0.01	20	34	58	$< 10^{-12}$
	WFR-theory	100	0.01	26	42	68	$< 10^{-12}$
smooth log-cosh	FR-only	1000	0.001	23	31	44	$< 10^{-12}$
	W-only	1000	0.001	1680	2560	3813	$< 10^{-12}$
	WFR-adaptive	1000	0.001	28	44	70	$< 10^{-12}$
	WFR-fixed	1000	0.001	30	46	72	$< 10^{-12}$
	WFR-theory	1000	0.001	36	52	80	$< 10^{-12}$
smooth log-cosh	FR-only	1000	0.01	17	25	39	$< 10^{-12}$
	W-only	1000	0.01	1680	2560	3813	$< 10^{-12}$
	WFR-adaptive	1000	0.01	26	42	68	$< 10^{-12}$
	WFR-fixed	1000	0.01	28	44	70	$< 10^{-12}$
	WFR-theory	1000	0.01	26	42	68	$< 10^{-12}$

Table 4: Schedule sweep for WFR-fixed: fraction c, transport $h = c/\beta$, expectation batches to gap $< 10^{-3}$, and final gap.

target	(Λ, ε)	c	$h = c/\beta$	$b_{10^{-3}}$	$\mathrm{gap}_{\mathrm{final}}$
Gaussian	$(100, 0.001)$	0	0.0000	36	0
		0.05	0.0500	44	0
		0.1	0.1000	40	0
		0.25	0.2500	36	0
		0.5	0.5000	32	0
		0.9	0.9000	30	0
Gaussian	$(100, 0.01)$	0	0.0000	30	0
		0.05	0.0500	44	0
		0.1	0.1000	40	0
		0.25	0.2500	36	0
		0.5	0.5000	32	0
		0.9	0.9000	30	0
Gaussian	$(1000, 0.001)$	0	0.0000	41	0
		0.05	0.0500	56	0
		0.1	0.1000	52	0
		0.25	0.2500	48	0
		0.5	0.5000	44	$< 10^{-12}$
		0.9	0.9000	42	$< 10^{-12}$
Gaussian	$(1000, 0.01)$	0	0.0000	36	0
		0.05	0.0500	56	0
		0.1	0.1000	52	0
		0.25	0.2500	48	0
		0.5	0.5000	44	$< 10^{-12}$
		0.9	0.9000	42	$< 10^{-12}$
smooth log-cosh	$(100, 0.001)$	0	0.0000	31	$< 10^{-12}$
		0.05	0.0049	46	$< 10^{-12}$
		0.1	0.0098	42	$< 10^{-12}$
		0.25	0.0246	38	$< 10^{-12}$
		0.5	0.0492	34	$< 10^{-12}$
		0.9	0.0886	30	$< 10^{-12}$
smooth log-cosh	$(100, 0.01)$	0	0.0000	25	$< 10^{-12}$
		0.05	0.0049	44	$< 10^{-12}$
		0.1	0.0098	42	$< 10^{-12}$
		0.25	0.0246	38	$< 10^{-12}$
		0.5	0.0492	34	$< 10^{-12}$
		0.9	0.0886	30	$< 10^{-12}$
smooth log-cosh	$(1000, 0.001)$	0	0.0000	31	$< 10^{-12}$
		0.05	0.0005	56	$< 10^{-12}$
		0.1	0.0010	52	$< 10^{-12}$
		0.25	0.0025	48	$< 10^{-12}$
		0.5	0.0050	46	$< 10^{-12}$
		0.9	0.0090	42	$< 10^{-12}$
smooth log-cosh	$(1000, 0.01)$	0	0.0000	25	$< 10^{-12}$
		0.05	0.0005	50	$< 10^{-12}$
		0.1	0.0010	48	$< 10^{-12}$
		0.25	0.0025	46	$< 10^{-12}$
		0.5	0.0050	44	$< 10^{-12}$
		0.9	0.0090	42	$< 10^{-12}$